PROJECT SYNOPSIS

CIVIC REPORTER

An AI-Assisted, Geotagged, Multilingual Civic Issue Reporting and Resolution Management System with Crowd-Validated Prioritization and Worker Accountability Tracking

Domain: Smart City Governance / E-Governance / Applied Artificial Intelligence

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1. PROJECT TITLE

Civic Reporter: An AI-Assisted, Geotagged, Multilingual Civic Issue Reporting and Resolution Management System with Crowd-Validated Prioritization and Worker Accountability Tracking.

2. ABSTRACT

Rapid urbanization has placed increasing strain on municipal bodies responsible for maintaining civic infrastructure, and citizens routinely face potholes, garbage accumulation, broken streetlights, water-supply disruptions, and air-quality concerns that go unreported or unresolved due to slow, opaque, and paper-driven grievance mechanisms. Civic Reporter is a two-panel web application — a User Panel and an Admin Panel — designed to close this gap by combining computer vision, generative AI, geolocation, and crowd-sourced validation into a single accountable workflow.

In the current, already-implemented version of the system, a registered citizen photographs an issue directly through the application's geotag-enabled camera; the captured image automatically embeds precise location metadata. The image is uploaded and processed through the Cloudinary API, which identifies the category of the issue (pothole, garbage, broken streetlight, water leakage, air-quality hazard, and similar categories). The Google Gemini API then drafts a formal, department-ready complaint letter from the classified issue and the location data, and this letter can be regenerated in 13 or more Indian and international languages so that citizens are not restricted by their comfort with English. Gemini is further used to infer the correct receiving authority for the complaint — for example, a Municipal Corporation, a Gram Panchayat, or another relevant local body — based on the location and issue type. The complaint location is plotted on an embedded Google Map, and other citizens in the vicinity or across the platform can upvote an existing complaint instead of filing a duplicate, which raises its visible priority. Upon submission, the citizen receives a unique Complaint ID for tracking and an automated confirmation email; a second automated email, containing the before-and-after photographs of the issue, is sent once the complaint is marked resolved.

The Admin Panel allows municipal staff to review incoming complaints, change their status (Pending, In Progress, Resolved, Declined, or Not True/Invalid), register new field workers, and

assign a specific worker to a specific complaint, creating a clear chain of accountability from citizen report to on-ground resolution.

Building on this working foundation, the project's next phase focuses on removing the platform's dependency on third-party, paid AI services for image classification by training a custom Convolutional Neural Network (CNN) on a locally curated civic-issue dataset; introducing dedicated, department-wise complaint queues (Water, Air, Garbage, Street/Streetlight, Potholes, and other categories) so that each department only sees relevant complaints; and adding day-by-day, percentage-based resolution progress tracking so that citizens can see how their complaint is progressing rather than only its final status. Collectively, these enhancements aim to deliver a transparent, cost-efficient, and highly accountable e-governance platform that shortens resolution times, improves resource allocation for municipal bodies, and increases citizen trust and participation.

3. INTRODUCTION / BACKGROUND

Effective management of civic infrastructure is a cornerstone of smart-city development. As urban populations grow, municipal corporations and local self-government bodies (such as Gram Panchayats and Nagar Panchayats) face increasing pressure to maintain public amenities and respond to citizen grievances quickly and fairly. Historically, reporting a civic issue has meant a physical visit to a municipal office, a phone call to an overloaded helpline, or, at best, a generic web form that captures a free-text description with no location verification and no way to independently confirm that the described issue actually exists.

The widespread availability of smartphones with built-in cameras and GPS sensors, combined with the recent maturity of cloud-based computer vision and large language models such as Google Gemini, has created a practical opportunity to modernize this workflow end to end: from the moment a citizen notices a problem to the moment a worker closes it out. Civic Reporter is positioned within the broader domain of Smart City Governance, and its objective is to bridge the communication gap between citizens and local authorities through an automated, verifiable, and multilingual digital platform that is usable by citizens across literacy levels, language backgrounds, and technical comfort.

Unlike a purely academic proof-of-concept, Civic Reporter already exists as a functioning application with a live User Panel and Admin Panel; this synopsis documents both the system as currently built and the roadmap for its next development phase, which focuses on cost reduction, departmental scalability, and finer-grained transparency for citizens.

4. PROBLEM STATEMENT

There is no widely accessible, unified platform that allows citizens to report civic issues with verified location evidence, receive a professionally drafted and correctly routed complaint in their own language, see the complaint's real-world priority reflected through community validation, and track its resolution transparently — while giving municipal administrators the tools to manage complaints, register and assign field workers, and maintain accountability for on-ground resolution. The absence of such a system results in misrouted or duplicated complaints, delayed resolutions, poor visibility for citizens, and inefficient allocation of municipal workforce.

5. OBJECTIVES

1. To develop a citizen-facing User Panel with a geotag-enabled in-app camera that captures civic issue photographs and automatically embeds accurate GPS location metadata at the moment of capture.
2. To automatically classify the captured issue image (pothole, garbage, broken streetlight, water leakage, air-quality concern, and other categories) using the Cloudinary API in the current implementation, and to progressively replace this with a custom-trained image recognition model to remove dependency on external paid APIs.
3. To use the Google Gemini API to auto-generate a formal, department-ready complaint letter from the classified issue and location data, and to allow this letter to be regenerated in 13 or more languages for accessibility across linguistic groups.
4. To use the Google Gemini API to infer and recommend the correct receiving authority for a complaint (Municipal Corporation, Gram Panchayat, or other relevant local body) based on the issue's location and category.
5. To visually plot each complaint's location on an integrated Google Map and enable a community-driven upvoting mechanism so that citizens can endorse existing complaints, increasing their visible priority instead of creating duplicate entries.
6. To generate a unique Complaint ID for every submission and provide citizens with a self-service tracking mechanism, supported by automated email notifications at submission and at resolution (including before-and-after photographic evidence).

7. To build an Admin Panel that allows municipal staff to update complaint status (Pending, In Progress, Resolved, Declined, Not True), register new field workers, and assign specific workers to specific complaints for on-ground resolution.
8. To extend the platform, in its next development phase, with department-wise complaint segregation (Water, Air, Garbage, Street/Streetlight, Potholes, and other categories) and day-by-day, percentage-based resolution progress tracking visible to the citizen.

6. MOTIVATION AND NEED

Municipal grievance redressal in most Indian cities and towns still suffers from high latency, inconsistent categorization, and a near-total lack of visibility for the citizen once a complaint is filed. A resident who reports a pothole by phone or through a generic web form typically has no way to know whether the complaint reached the correct department, whether it has been assigned to anyone, or when it might be resolved. This opacity erodes trust and discourages future reporting, even though the underlying issues remain unresolved.

Language is a further, often overlooked, barrier: formal municipal complaint formats are usually expected in English or in the state's official administrative language, which excludes a large share of citizens who are more comfortable in their regional or local language. By using generative AI to draft the complaint letter and offer it in 13 or more languages, Civic Reporter removes this barrier without requiring the citizen to know the correct bureaucratic phrasing or the correct receiving department — both of which the system infers automatically.

On the administrative side, municipal bodies frequently struggle to track which worker is handling which issue and how far along that work has progressed, leading to duplicated effort, missed complaints, and no institutional memory of resolution quality. A structured worker-registration and assignment module, combined with status tracking, gives administrators a single source of truth for field operations.

Finally, relying indefinitely on third-party, paid AI services for image classification introduces recurring operational cost and a dependency that is unsuitable for a system intended to scale to many municipalities. Training a custom, domain-specific image recognition model — tailored to the specific visual signatures of potholes, garbage accumulation, streetlight damage, and similar issues in the local context — is therefore both a cost-control measure and an opportunity to improve classification accuracy for cases that generic vision APIs are not specifically tuned for.

7. EXISTING SOLUTIONS / BRIEF LITERATURE SURVEY

The application of Information and Communication Technology (ICT) to civic-issue management has been studied extensively, and the literature broadly clusters around four themes that are directly relevant to Civic Reporter: automated visual classification of civic issues, geotagged and crowd-sourced reporting, AI-assisted routing and drafting of complaints, and field-worker tracking.

Sharma and Kumar (1) proposed a CNN-based pothole detection system for smart cities and demonstrated strong classification accuracy, but their work addressed a single issue category and did not extend to citizen-facing reporting, multi-department routing, or worker assignment. Chen et al. (3) similarly showed the effectiveness of custom-trained computer vision models for automated solid-waste classification, again as a narrow, single-domain solution rather than a holistic civic portal. Wang et al. (6) extended visual damage detection to broader urban infrastructure using YOLO-based object detection, reinforcing that custom, purpose-trained vision models can outperform generic classifiers for domain-specific civic imagery — a finding that directly motivates Civic Reporter's planned move away from a general-purpose image API toward a custom-trained model.

On the reporting and routing side, Kim and Lee (8) explored AI-assisted routing of complaints using image analysis together with natural language processing, but their reliance on generalized, paid cloud AI services raised concerns about recurring cost, continuous connectivity requirements, and data-privacy exposure — concerns that Civic Reporter's roadmap for a self-trained model is specifically designed to mitigate over time. Reddy et al. (7) reviewed mobile crowdsensing approaches for smart-city applications and highlighted the value of citizen-generated data streams when they are properly geotagged and validated, which supports Civic Reporter's use of GPS-embedded photo capture and community upvoting as a lightweight, low-cost validation layer rather than relying solely on administrative verification.

Kumar et al. (9) emphasized that manual or self-reported location entry frequently results in unresolvable complaints due to spatial inaccuracy or ambiguous landmark descriptions, underscoring the importance of automatic, metadata-driven geotagging at the point of capture rather than asking citizens to describe or select a location themselves. Gupta and Singh (2) reviewed e-governance frameworks across developing regions and found that the absence of real-time status visibility is a primary driver of citizen distrust in digital grievance systems — a gap

that Civic Reporter addresses through its Complaint ID tracking and, in its next phase, percentage-based progress visibility.

On the operations side, Patel and Desai (4) developed a GPS- and IoT-based framework for tracking municipal field workers and found that real-time monitoring measurably improved task-completion rates and reduced resource mismanagement, but their system operated independently of any citizen-facing reporting mechanism. Al-Hader and Rodzi (5) discussed the value of decentralized, department-specific data processing within smart-city infrastructure, which supports Civic Reporter's planned segregation of complaints into distinct department queues (Water, Air, Garbage, Street, Potholes) rather than a single undifferentiated complaint pool. Zhang et al. (10) similarly showed, in the context of streetlight management, that isolating departmental data streams improves response times compared with centralized, undifferentiated systems.

Existing city grievance portals, summarized in the table below, typically automate only one part of this workflow — either classification, or worker tracking, or centralized form collection — and none of the reviewed systems combine automatic multilingual complaint drafting, AI-based authority identification, crowd-validated prioritization, geotagged evidence capture, department-wise routing, and worker-level accountability in a single citizen-facing platform. Civic Reporter's contribution is this synthesis: it draws on custom visual classification (1, 3, 6), geotagged crowdsourcing (7, 9), AI-assisted drafting and routing without a permanent dependency on paid APIs (8), department-specific queues (5, 10), and structured worker tracking (4), while adding two capabilities not addressed in the reviewed literature — automatic multilingual complaint generation and community upvoting as a lightweight prioritization signal.

Sr. No.	Source / Existing Solution	Approach	Key Features	Limitations
1	Sharma & Kumar (2021)	CNN for pothole detection	High-accuracy single-category image classification	Limited to potholes; no citizen reporting or worker tracking
2	Patel & Desai (2023)	GPS/IoT tracking for workers	Real-time field-worker tracking	Not integrated with citizen complaint intake
3	Kim & Lee (2021)	NLP + image analysis API	Automated complaint routing	Relies on paid third-party AI APIs; no multilingual support

4	Typical Municipal Web Portals	Manual web forms	Centralized complaint collection	No AI categorization, no geotag verification, no live tracking

8. PROPOSED SOLUTION

8.1 User Panel Workflow

A citizen registers and logs in to the User Panel, then uses the application's built-in geotag-enabled camera to photograph the issue; the photo's metadata automatically captures precise GPS coordinates at the moment of capture, removing any need for the citizen to manually enter or describe a location. The image is uploaded and passed to the Cloudinary API, which identifies the issue category (for example, Pothole, Garbage, Broken Streetlight, Water Leakage, or Air-Quality Hazard). The Google Gemini API then drafts a formal complaint letter using the classified category and the location data, and the citizen can view or regenerate this letter in any of 13 or more supported languages. Gemini also analyzes the location and issue category to recommend the correct receiving authority — such as a Municipal Corporation or Gram Panchayat — so the complaint is addressed appropriately without the citizen needing prior knowledge of local administrative jurisdictions.

The complaint's location is displayed on an embedded Google Map, giving both the citizen and, later, the assigned worker an unambiguous visual reference. Other citizens can upvote an existing nearby complaint rather than submitting a duplicate, which increases the complaint's visible priority and signals to administrators that an issue affects multiple residents. Upon submission, the citizen receives a unique Complaint ID that can be used to track status at any time, along with an automated confirmation email. When the complaint is marked resolved by the Admin Panel, the citizen automatically receives a second email containing the before-and-after photographs of the issue as visual proof of resolution.

8.2 Admin Panel Workflow

Municipal administrators log in to a separate Admin Panel where all incoming complaints are visible, sortable, and filterable. Administrators can update a complaint's status — Pending, In Progress, Resolved, Declined, or Not True/Invalid (for cases such as duplicate or fabricated reports) — and can register new field workers directly through the panel, capturing their contact details and department. Once a worker is registered, administrators can assign that worker to a

specific complaint, creating a traceable link between a citizen's report and the individual responsible for resolving it.

8.3 Planned Phase-II Enhancements

The next development phase focuses on three enhancements. First, the Cloudinary-based image classification will be progressively replaced with a custom-trained Convolutional Neural Network built on a locally curated dataset of civic-issue photographs, removing the platform's dependency on external, paid AI services for this step and improving accuracy for issue types that are specific to the deployment region. Second, complaints will be segregated into distinct department-wise queues — Water, Air, Garbage, Street/Streetlight, Potholes, and any additional categories identified during data collection — so that each department's dashboard shows only its relevant complaints, mirroring how real municipal bodies are organizationally structured. Third, the citizen-facing tracking experience will be upgraded from a status label to a day-by-day, percentage-based progress indicator (for example, 25% Assigned, 50% In Progress, 100% Resolved), giving citizens finer-grained visibility into how their complaint is moving through the resolution pipeline.

9. PROPOSED METHODOLOGY / APPROACH

9. Requirements Consolidation: Formalizing the issue categories, department structures, language list, and administrative workflows already validated in the current version of Civic Reporter, and defining the additional requirements for department segregation and percentage tracking.
10. Dataset Collection and Curation: Assembling a labeled image dataset covering the target issue categories (potholes, garbage, broken streetlights, water leakage, air-quality indicators, and others) from field photography, open civic-issue datasets, and images already collected through the live application, with attention to varied lighting, weather, and camera-angle conditions.
11. Custom Model Training: Designing and training a Convolutional Neural Network (or fine-tuning a suitable pretrained vision backbone) for multi-class civic-issue classification, with data augmentation to improve robustness, followed by iterative evaluation against a held-out test set.
12. System and Database Redesign: Extending the existing database schema to support department-wise complaint queues and granular, percentage-based status tracking, and updating the Admin Panel's data model accordingly.

13. Development and Integration: Implementing the department-specific dashboards, the percentage-tracking UI for citizens, and integrating the custom-trained model into the existing image-upload pipeline in place of the Cloudinary classification call.
14. Testing: Conducting functional testing of the new department routing logic, geolocation accuracy testing, and precision/recall evaluation of the custom AI model against the previous Cloudinary-based classification baseline.
15. Evaluation and Refinement: Collecting user and administrator feedback on the enhanced system and refining classification thresholds, department rules, and the progress-tracking granularity accordingly.

10. PROPOSED TECHNOLOGY / TOOLS

Component	Technology / Tool	Justification
Frontend (User & Admin Panels)	React.js (Vite)	Fast, component-driven, responsive interface shared across the citizen and admin experiences.
Backend / API Layer	Node.js with Express.js, or Firebase Cloud Functions	Scalable, non-blocking handling of concurrent complaint submissions, notifications, and admin operations.
Database	Firebase Firestore / MongoDB	Flexible, document-oriented schema well suited to varying complaint fields, geotags, votes, and worker records.
Authentication	Firebase Authentication	Secure citizen and admin login with minimal custom auth infrastructure to maintain.
Current Image Classification	Cloudinary API	Existing integration used for automatic issue-category detection from uploaded photographs.
Planned Custom AI Model	Python, TensorFlow / PyTorch (CNN)	Removes recurring third-party API cost and improves accuracy for locally relevant civic-issue imagery.
Complaint Drafting & Authority Detection	Google Gemini API	Generates formal, multilingual complaint letters and infers the correct receiving municipal authority.
Geolocation & Mapping	HTML5 Geolocation API, EXIF metadata extraction, Google Maps API	Accurate GPS capture at the point of photography and visual plotting of complaint locations.
Notifications	Email service (e.g., Nodemailer / SendGrid) via backend triggers	Automated submission and resolution emails, including before-and-after images.
Hosting / Deployment	Vercel (frontend) with a managed backend host	Consistent with the platform's existing live deployment approach.

Note: the technology choices above reflect the stack already in use for the live application and standard extensions of it; they can be adjusted to match the exact backend and hosting configuration finalized for the project report.

11. EXPECTED OUTCOMES / DELIVERABLES

Deliverable	Description
User Panel	Citizen-facing application with geotag camera capture, AI-drafted multilingual complaint letters, authority detection, map-based location display, upvoting, and Complaint ID tracking.
Admin Panel	Administrative dashboard for status management, worker registration, and worker-to-complaint assignment.
Custom AI Model	A self-trained image classification model covering the target civic-issue categories, replacing the current third-party classification API.
Department-Wise Queues	Segregated complaint views for Water, Air, Garbage, Street/Streetlight, Potholes, and additional categories as identified.
Percentage-Based Tracking	Day-by-day resolution progress indicator visible to citizens, replacing or supplementing the current status labels.
Notification System	Automated submission and resolution emails, the latter including before-and-after photographic evidence.
Project Report & Documentation	Complete technical documentation covering architecture, model training methodology, and evaluation results.

12. PRELIMINARY EVALUATION PLAN

The enhanced system will be evaluated along three dimensions.

- AI Model Performance: The custom-trained image classification model will be measured using Accuracy, Precision, Recall, and F1-Score on a held-out test dataset, and its results will be compared directly against the existing Cloudinary-based classification baseline for the same test images.
- Functional Testing: Verification that geotagging captures coordinates within an acceptable accuracy radius (target: within 5 meters), that complaints are routed to the correct department queue, and that Gemini-generated complaint letters correctly identify the receiving authority for a sample of known jurisdictions.
- Usability and Workflow Testing: Structured user testing to confirm that a citizen can report an issue in a small number of steps, that the multilingual letter generation is understandable to native speakers of the tested languages, and that the percentage-based tracking updates accurately reflect worker-reported progress.

13. FEASIBILITY ANALYSIS

Technical Feasibility: High. The core system is already implemented and live, and the proposed enhancements (custom CNN training, department segregation, percentage tracking) use well-documented, widely supported tools and standard database-schema extensions.

Resource Feasibility: High. Open civic-issue image datasets, combined with images already collected through the live application, provide a practical starting point for training the custom model without requiring large-scale new data collection.

Skill Feasibility: The enhancement work aligns with standard full-stack development and applied machine-learning skills, both of which are already reflected in the existing, functioning implementation.

Time Feasibility: Achievable within a standard academic timeline, provided the model-training phase is started early enough to allow for iterative evaluation and refinement.

Risk: The primary risk is achieving classification accuracy from the custom model that matches or exceeds the current Cloudinary-based baseline across varied lighting, weather, and image-quality conditions. This will be mitigated through data augmentation, a sufficiently diverse training dataset, and a phased rollout in which the custom model runs alongside the existing API before fully replacing it.

14. PROJECT SCOPE

14.1 Already Implemented (Current System)

- User registration/login and a geotag-enabled in-app camera for photo capture.
- Automatic issue classification via the Cloudinary API.
- AI-generated, multilingual (13+ languages) complaint letter drafting via the Google Gemini API.
- AI-based identification of the correct receiving municipal authority.
- Google Maps display of complaint location.
- Community upvoting to increase complaint priority.
- Unique Complaint ID generation and citizen-side tracking.
- Automated submission and resolution emails, the latter with before-and-after images.

- Admin Panel for status management (Solved, In Progress, Declined, Not True) and worker registration and assignment.

14.2 In Scope for Next Phase

- Training and integrating a custom image recognition model to replace the Cloudinary dependency.
- Department-wise segregation of complaints (Water, Air, Garbage, Street/Streetlight, Potholes, and additional categories).
- Day-by-day, percentage-based resolution progress tracking visible to citizens.

14.3 Out of Scope

- Handling of emergency services (Police, Fire, Ambulance).
- Automated physical repair, dispatch of autonomous drones, or robotic remediation.
- Financial transactions, billing, fines, or municipal tax collection.
- Voice-based complaint registration (not part of the current roadmap).

15. TENTATIVE TIMELINE

Phase	Activities	Duration
Phase 1	Requirements consolidation and dataset planning for the custom AI model and department segregation	Weeks 1–2
Phase 2	Dataset collection/curation and database schema redesign for department queues and percentage tracking	Weeks 3–5
Phase 3	Custom CNN model training and iterative evaluation against the existing baseline	Weeks 6–9
Phase 4	Backend integration of the trained model, department-wise dashboards, and percentage-tracking UI	Weeks 10–12
Phase 5	System testing — functional, geolocation, and AI model evaluation	Weeks 13–14
Phase 6	Refinement, final documentation, and report/presentation preparation	Weeks 15–16

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